Example 1: Sweet Apple Acres

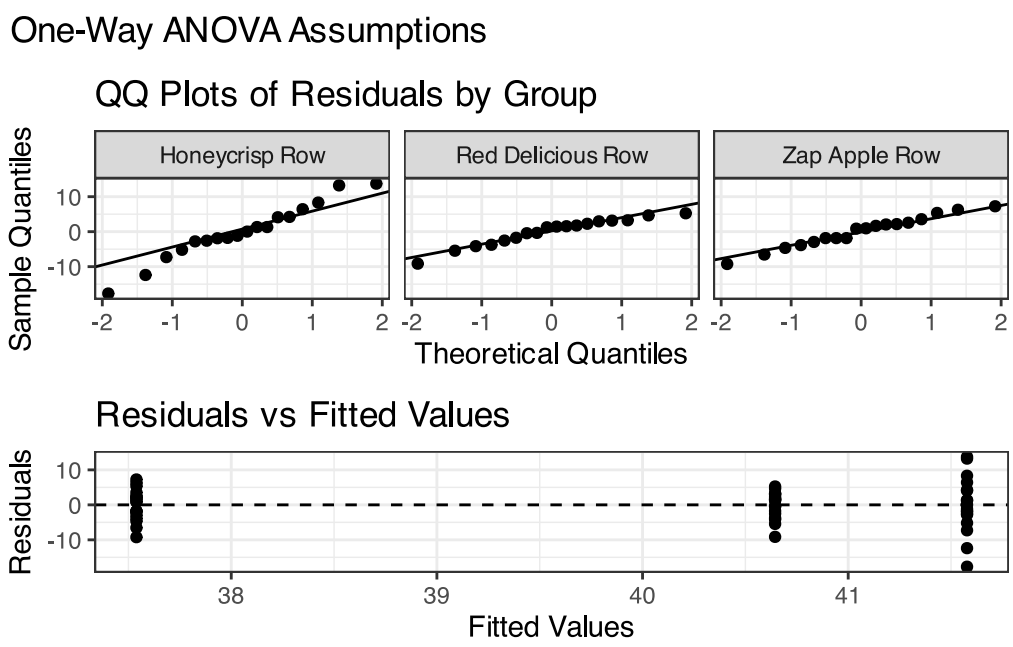
Applejack has come to us: "Between the Red Delicious row, the Honeycrisp row, and that new Zap Apple row Granny Smith insisted on plantin', I reckon one of 'em oughta be doin' better than the others. Can you tell me if they're really any different, or am I just imaginin' things?"

- Glimpse of data:

```
# A tibble: 3 × 2
  row               bushels
  <chr>               <dbl>
1 Red Delicious Row    42.2
2 Red Delicious Row    45.3
3 Red Delicious Row    42.1
```

- Checking the normality of our Sweet Apple Acres dataset,

```
apple_harvest |> anova_check(outcome = bushels,
                              grouping = row)
```



- Constructing the ANOVA table,

```
apple_harvest |> anovas(outcome = bushels,
                         grouping = row)
```

```
One-way ANOVA
```

```
Source     df         SS       MS      F  p-value
──────────────────────────────────────────────────
row         2   160.9693  80.4846  2.468    0.095
Residuals  51  1663.4383  32.6164
──────────────────────────────────────────────────


Hypotheses (row):
H₀: All group means are equal (no effect of row)
H₁: At least one group mean differs (effect of row exists)
Test statistic: F(2, 51) = 2.468
p-value: 0.095
Conclusion: Fail to reject the null hypothesis (p = 0.095 ≥ α = 0.050)
```

- Applying the Kruskal-Wallis,

```
apple_harvest |> kruskal_wallis(outcome = bushels,
                                grouping = row)
```

```
Kruskal-Wallis test

M[Honeycrisp Row] = 41 (n = 18)
M[Red Delicious Row] = 42 (n = 18)
M[Zap Apple Row] = 38.45 (n = 18)

Hypotheses:
H₀: All group distributions are equal (no effect of row)
H₁: At least one group distribution differs in location (effect of row exists)
Test statistic: H(2) = 5.953
p-value: 0.051
Conclusion: Fail to reject the null hypothesis (p = 0.051 ≥ α = 0.050)
```

Q: Should we perform posthoc testing?

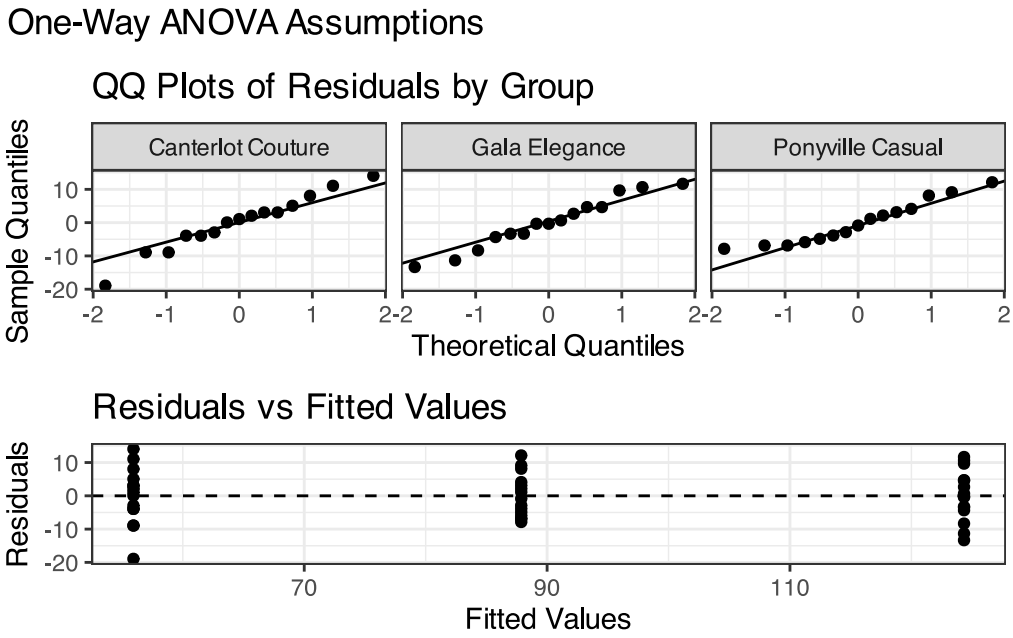
Example 2: Rarity's Boutique

- Rarity flits between racks of fabric: "Darling, I need your professional opinion. Between my Canterlot Couture line, my everyday Ponyville Casual line, and my absolutely *dazzling* Gala Elegance line, I use wildly different amounts of gemstones. Is the difference really as dramatic as I think?"

```
# A tibble: 3 × 2
  collection        gem_count
  <chr>                 <dbl>
1 Canterlot Couture        59
2 Canterlot Couture        57
3 Canterlot Couture        64
```

- Checking the normality of Rarity's dataset,

```
boutique_gems |> anova_check(outcome = gem_count,
                              grouping = collection)
```



- Constructing the ANOVA table,

```
boutique_gems |> anovas(outcome = gem_count,
                        grouping = collection)
```

```
One-way ANOVA
```

```
Source      df          SS          MS        F  p-value
──────────────────────────────────────────────────────────
collection   2  35140.5778  17570.2889  311.110  < 0.001
Residuals   42   2372.0000     56.4762
──────────────────────────────────────────────────────────


Hypotheses (collection):
H₀: All group means are equal (no effect of collection)
H₁: At least one group mean differs (effect of collection exists)
Test statistic: F(2, 42) = 311.11
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- Applying the Kruskal-Wallis,

```
boutique_gems |> kruskal_wallis(outcome = gem_count,
                                grouping = collection)
```

```
Kruskal-Wallis test

M[Canterlot Couture] = 57 (n = 15)
M[Gala Elegance] = 124 (n = 15)
M[Ponyville Casual] = 87 (n = 15)

Hypotheses:
H₀: All group distributions are equal (no effect of collection)
H₁: At least one group distribution differs in location (effect of collection
exists)
Test statistic: H(2) = 39.148
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

Q: Should we perform posthoc testing?

Q: Treat this as an exploratory study and use the following output to determine which pairwise differences exist. Circle the output you are interpreting.

- Posthoc tests:

```
boutique_gems |> posthoc(outcome = gem_count,
                         grouping = collection,
                         adjust = TRUE,
                         nonparametric = FALSE)
```

```
Post-hoc analysis: Tukey's HSD

Following: one-way ANOVA (gem_count ~ collection)

Comparison                              Difference  p-value
──────────────────────────────────────────────────────────────
Gala Elegance vs Canterlot Couture         68.4000  < 0.001  *
Ponyville Casual vs Canterlot Couture      31.9333  < 0.001  *
Ponyville Casual vs Gala Elegance         -36.4667  < 0.001  *
──────────────────────────────────────────────────────────────

* p < α = 0.050
```

```
boutique_gems |> posthoc(outcome = gem_count,
                         grouping = collection,
                         adjust = FALSE,
                         nonparametric = FALSE)
```

```
Post-hoc analysis: Fisher's LSD

Following: one-way ANOVA (gem_count ~ collection)

Comparison                              Difference  p-value
──────────────────────────────────────────────────────────────
Gala Elegance vs Canterlot Couture         68.4000  < 0.001  *
Ponyville Casual vs Canterlot Couture      31.9333  < 0.001  *
Ponyville Casual vs Gala Elegance         -36.4667  < 0.001  *
──────────────────────────────────────────────────────────────

* p < α = 0.050
```

```
boutique_gems |> posthoc(outcome = gem_count,
                         grouping = collection,
                         adjust = TRUE,
                         nonparametric = TRUE)
```

```
Post-hoc analysis: Dunn's test (Bonferroni adjustment)

Following: Kruskal-Wallis (gem_count ~ collection)

Comparison                                    z  p-value
──────────────────────────────────────────────────────────
Canterlot Couture vs Gala Elegance     -6.257  < 0.001  *
Canterlot Couture vs Ponyville Casual  -3.128    0.003  *
Gala Elegance vs Ponyville Casual       3.128    0.003  *
──────────────────────────────────────────────────────────

* p < α = 0.050
```

```
boutique_gems |> posthoc(outcome = gem_count,
                         grouping = collection,
                         adjust = FALSE,
                         nonparametric = TRUE)
```

```
Post-hoc analysis: Dunn's test (no adjustment)

Following: Kruskal-Wallis (gem_count ~ collection)

Comparison                                    z  p-value
──────────────────────────────────────────────────────────
Canterlot Couture vs Gala Elegance     -6.257  < 0.001  *
Canterlot Couture vs Ponyville Casual  -3.128  < 0.001  *
Gala Elegance vs Ponyville Casual       3.128  < 0.001  *
──────────────────────────────────────────────────────────

* p < α = 0.050
```

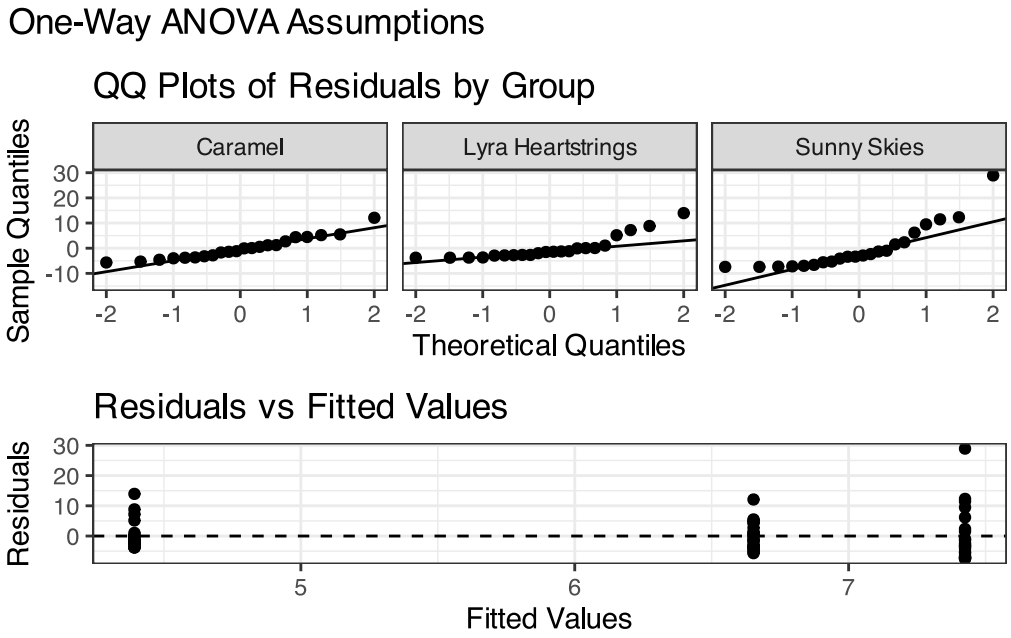
Example 3: Ponyville Diner

- Sunny Skies wipes down the counter after the lunch rush: "Feels like Caramel always walks away with better tips than me and Lyra put together. Before I start complainin' to the manager, can you check if there's actually a difference in how much we're bringing in?"

```
# A tibble: 3 × 2
  server      tip_bits
  <chr>          <dbl>
1 Sunny Skies     4.08
2 Sunny Skies     6.12
3 Sunny Skies     0.12
```

- Checking the normality of the Ponyville Diner dataset,

```
ponyville_diner_tips |> anova_check(outcome = tip_bits,
                                    grouping = server)
```



- Constructing the ANOVA table,

```
ponyville_diner_tips |> anovas(outcome = tip_bits,
                               grouping = server)
```

```
One-way ANOVA

Source     df         SS       MS      F  p-value
```

```
──────────────────────────────────────────────────
server      2   109.2923  54.6461  1.356    0.265
Residuals  63  2539.3768  40.3076
──────────────────────────────────────────────────


Hypotheses (server):
H₀: All group means are equal (no effect of server)
H₁: At least one group mean differs (effect of server exists)
Test statistic: F(2, 63) = 1.356
p-value: 0.265
Conclusion: Fail to reject the null hypothesis (p = 0.265 ≥ α = 0.050)
```

- Applying the Kruskal-Wallis,

```
ponyville_diner_tips |> kruskal_wallis(outcome = tip_bits,
                                       grouping = server)
```

```
Kruskal-Wallis test

M[Caramel] = 6.03 (n = 22)
M[Lyra Heartstrings] = 2.94 (n = 22)
M[Sunny Skies] = 4.305 (n = 22)

Hypotheses:
H₀: All group distributions are equal (no effect of server)
H₁: At least one group distribution differs in location (effect of server exists)
Test statistic: H(2) = 4.235
p-value: 0.120
Conclusion: Fail to reject the null hypothesis (p = 0.120 ≥ α = 0.050)
```

Q: Should we perform posthoc testing?

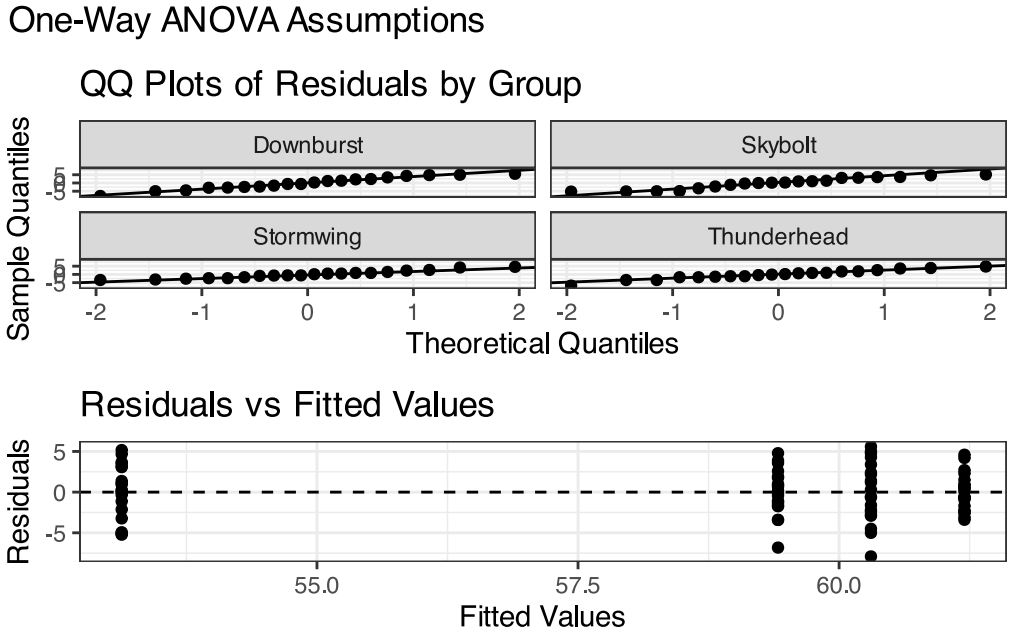
Let's Practice! 1: Wonderbolts Academy

- Fleetfoot drops a stack of stopwatch printouts on the table: "Command wants to know if lap times actually differ across our four training squads: Thunderhead, Stormwing, Skybolt, and Downburst."

```
# A tibble: 3 × 2
  squad       lap_seconds
  <chr>             <dbl>
1 Thunderhead        60.4
2 Thunderhead        59.6
3 Thunderhead        62
```

- Checking the normality,

```
wonderbolts_laptimes |> anova_check(outcome = lap_seconds,
                                    grouping = squad)
```



- Constructing the ANOVA table,

```
wonderbolts_laptimes |> anovas(outcome = lap_seconds,
                               grouping = squad)
```

```
One-way ANOVA

Source     df        SS        MS       F  p-value
```

```
──────────────────────────────────────────────────
squad       3  804.4305  268.1435  28.098  < 0.001
Residuals  76  725.2770    9.5431
──────────────────────────────────────────────────


Hypotheses (squad):
H₀: All group means are equal (no effect of squad)
H₁: At least one group mean differs (effect of squad exists)
Test statistic: F(3, 76) = 28.098
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- Applying the Kruskal-Wallis,

```
wonderbolts_laptimes |> kruskal_wallis(outcome = lap_seconds,
                                       grouping = squad)
```

```
Kruskal-Wallis test

M[Downburst] = 60.15 (n = 20)
M[Skybolt] = 53.35 (n = 20)
M[Stormwing] = 60.95 (n = 20)
M[Thunderhead] = 59.45 (n = 20)

Hypotheses:
H₀: All group distributions are equal (no effect of squad)
H₁: At least one group distribution differs in location (effect of squad exists)
Test statistic: H(3) = 38.039
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

Q: Should we perform posthoc testing?

Q: Treat this as a confirmatory study and use the following output to determine which pairwise differences exist. Circle the output you are interpreting.

- Posthoc tests:

```
wonderbolts_laptimes |> posthoc(outcome = lap_seconds,
                                grouping = squad,
                                adjust = TRUE,
                                nonparametric = FALSE)
```

```
Post-hoc analysis: Tukey's HSD

Following: one-way ANOVA (lap_seconds ~ squad)

Comparison                Difference  p-value
──────────────────────────────────────────────
Skybolt vs Downburst         -7.1750  < 0.001  *
Stormwing vs Downburst        0.8950    0.796
Thunderhead vs Downburst     -0.8900    0.799
Stormwing vs Skybolt          8.0700  < 0.001  *
Thunderhead vs Skybolt        6.2850  < 0.001  *
Thunderhead vs Stormwing     -1.7850    0.269
──────────────────────────────────────────────

* p < α = 0.050
```

```
wonderbolts_laptimes |> posthoc(outcome = lap_seconds,
                                grouping = squad,
                                adjust = FALSE,
                                nonparametric = FALSE)
```

```
Post-hoc analysis: Fisher's LSD

Following: one-way ANOVA (lap_seconds ~ squad)

Comparison                Difference  p-value
──────────────────────────────────────────────
Skybolt vs Downburst         -7.1750  < 0.001  *
Stormwing vs Downburst        0.8950    0.362
Stormwing vs Skybolt          8.0700  < 0.001  *
Thunderhead vs Downburst     -0.8900    0.365
Thunderhead vs Skybolt        6.2850  < 0.001  *
Thunderhead vs Stormwing     -1.7850    0.072
──────────────────────────────────────────────

* p < α = 0.050
```

```
wonderbolts_laptimes |> posthoc(outcome = lap_seconds,
                                grouping = squad,
                                adjust = TRUE,
                                nonparametric = TRUE)
```

```
Post-hoc analysis: Dunn's test (Bonferroni adjustment)

Following: Kruskal-Wallis (lap_seconds ~ squad)

Comparison                      z  p-value
──────────────────────────────────────────
Downburst vs Skybolt        4.787  < 0.001  *
Downburst vs Stormwing     -0.915    1.000
Skybolt vs Stormwing       -5.703  < 0.001  *
Downburst vs Thunderhead    0.701    1.000
Skybolt vs Thunderhead     -4.086  < 0.001  *
Stormwing vs Thunderhead    1.616    0.318
──────────────────────────────────────────

* p < α = 0.050
```

```
wonderbolts_laptimes |> posthoc(outcome = lap_seconds,
                                grouping = squad,
                                adjust = FALSE,
                                nonparametric = TRUE)
```

```
Post-hoc analysis: Dunn's test (no adjustment)

Following: Kruskal-Wallis (lap_seconds ~ squad)

Comparison                      z  p-value
──────────────────────────────────────────
Downburst vs Skybolt        4.787  < 0.001  *
Downburst vs Stormwing     -0.915    0.180
Skybolt vs Stormwing       -5.703  < 0.001  *
Downburst vs Thunderhead    0.701    0.242
Skybolt vs Thunderhead     -4.086  < 0.001  *
Stormwing vs Thunderhead    1.616    0.053
──────────────────────────────────────────

* p < α = 0.050
```

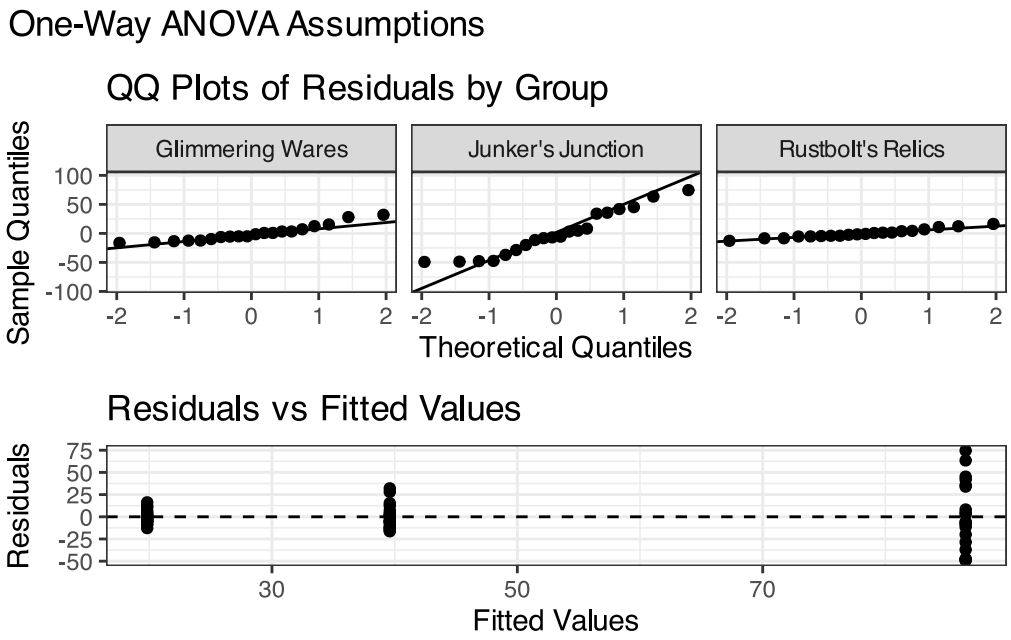
Let's Practice! 2: Klugetown Bazaar

- A weary trader waves you over between haggling matches: "Rustbolt, Glimmering, and Junker each swear their stall pulls in the best prices. I've got a season's worth of sale records — do the three stalls actually sell for different amounts, or is everyone just talkin' up their own wares?"

```
# A tibble: 3 × 2
  stall             sale_price_bits
  <chr>                       <dbl>
1 Rustbolt's Relics            21.2
2 Rustbolt's Relics            18.9
3 Rustbolt's Relics            26.8
```

- Checking the normality,

```
klugetown_bazaar |> anova_check(outcome = sale_price_bits,
                                grouping = stall)
```



- Constructing the ANOVA table,

```
klugetown_bazaar |> anovas(outcome = sale_price_bits,
                           grouping = stall)
```

```
One-way ANOVA

Source     df          SS          MS       F  p-value
```

```
──────────────────────────────────────────────────────
stall       2  46965.2768  23482.6384  40.982  < 0.001
Residuals  57  32661.2297    573.0040
──────────────────────────────────────────────────────


Hypotheses (stall):
H₀: All group means are equal (no effect of stall)
H₁: At least one group mean differs (effect of stall exists)
Test statistic: F(2, 57) = 40.982
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- Applying the Kruskal-Wallis,

```
klugetown_bazaar |> kruskal_wallis(outcome = sale_price_bits,
                                   grouping = stall)
```

```
Kruskal-Wallis test

M[Glimmering Wares] = 36.36 (n = 20)
M[Junker's Junction] = 80.29 (n = 20)
M[Rustbolt's Relics] = 18.565 (n = 20)

Hypotheses:
H₀: All group distributions are equal (no effect of stall)
H₁: At least one group distribution differs in location (effect of stall exists)
Test statistic: H(2) = 43.348
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

Q: Should we perform posthoc testing?

Q: Treat this as a confirmatory study and use the following output to determine which pairwise differences exist. Circle the output you are interpreting.

- Posthoc tests:

```
klugetown_bazaar |> posthoc(outcome = sale_price_bits,
                            grouping = stall,
                            adjust = TRUE,
                            nonparametric = FALSE)
```

```
Post-hoc analysis: Tukey's HSD

Following: one-way ANOVA (sale_price_bits ~ stall)

Comparison                                Difference  p-value
──────────────────────────────────────────────────────────────
Junker's Junction vs Glimmering Wares      46.9385  < 0.001  *
Rustbolt's Relics vs Glimmering Wares     -19.7740    0.030  *
Rustbolt's Relics vs Junker's Junction    -66.7125  < 0.001  *
──────────────────────────────────────────────────────────────

* p < α = 0.050
```

```
klugetown_bazaar |> posthoc(outcome = sale_price_bits,
                            grouping = stall,
                            adjust = FALSE,
                            nonparametric = FALSE)
```

```
Post-hoc analysis: Fisher's LSD

Following: one-way ANOVA (sale_price_bits ~ stall)

Comparison                                Difference  p-value
──────────────────────────────────────────────────────────────
Junker's Junction vs Glimmering Wares      46.9385  < 0.001  *
Rustbolt's Relics vs Glimmering Wares     -19.7740    0.011  *
Rustbolt's Relics vs Junker's Junction    -66.7125  < 0.001  *
──────────────────────────────────────────────────────────────

* p < α = 0.050
```

```
klugetown_bazaar |> posthoc(outcome = sale_price_bits,
                            grouping = stall,
                            adjust = TRUE,
                            nonparametric = TRUE)
```

```
Post-hoc analysis: Dunn's test (Bonferroni adjustment)

Following: Kruskal-Wallis (sale_price_bits ~ stall)

Comparison                                      z  p-value
──────────────────────────────────────────────────────────
Glimmering Wares vs Junker's Junction   -3.038    0.004  *
Glimmering Wares vs Rustbolt's Relics    3.540  < 0.001  *
Junker's Junction vs Rustbolt's Relics   6.578  < 0.001  *
──────────────────────────────────────────────────────────

* p < α = 0.050
```

```
klugetown_bazaar |> posthoc(outcome = sale_price_bits,
                            grouping = stall,
                            adjust = FALSE,
                            nonparametric = TRUE)
```

```
Post-hoc analysis: Dunn's test (no adjustment)

Following: Kruskal-Wallis (sale_price_bits ~ stall)

Comparison                                      z  p-value
──────────────────────────────────────────────────────────
Glimmering Wares vs Junker's Junction   -3.038    0.001  *
Glimmering Wares vs Rustbolt's Relics    3.540  < 0.001  *
Junker's Junction vs Rustbolt's Relics   6.578  < 0.001  *
──────────────────────────────────────────────────────────

* p < α = 0.050
```

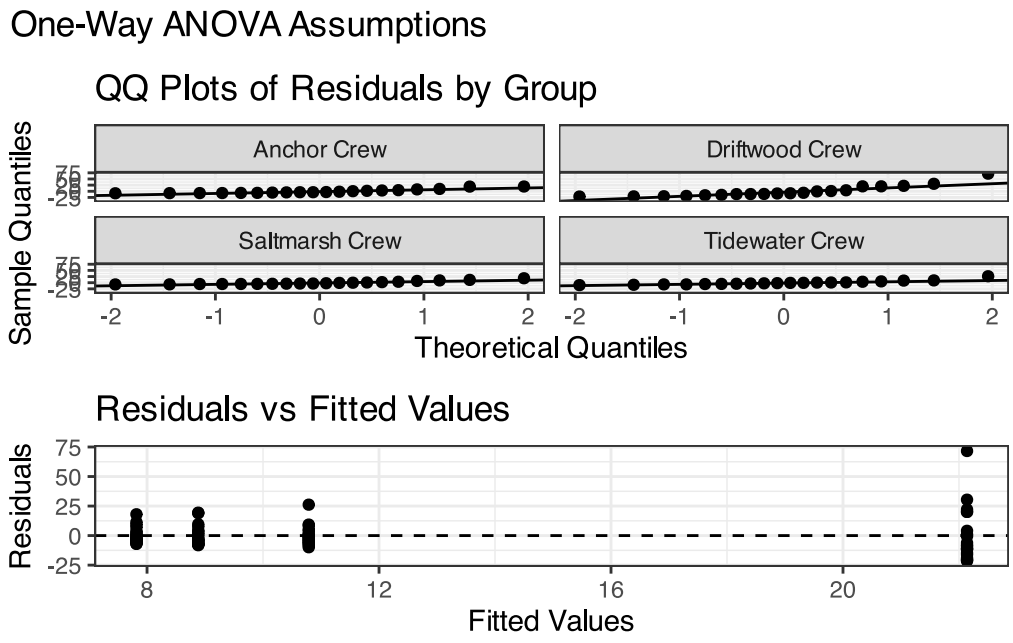
Let's Practice! 3: Baltimare Harbor

- The harbormaster spreads a stack of logbooks across the table: "Four dock crews unload every ship that comes through: Anchor, Tidewater, Saltmarsh, and Driftwood. Some of the sailors have been grumbling that one crew is dragging everyone's schedule behind. Before I say anything to the crews, I want to know if unloading time actually differs across them."

```
# A tibble: 3 × 2
  crew        unload_minutes
  <chr>                <dbl>
1 Anchor Crew           1.3
2 Anchor Crew           2.19
3 Anchor Crew           5.13
```

- Checking the normality,

```
baltimare_harbor |> anova_check(outcome = unload_minutes,
                                grouping = crew)
```



- Constructing the ANOVA table,

```
baltimare_harbor |> anovas(outcome = unload_minutes,
                           grouping = crew)
```

```
One-way ANOVA
```

```
Source     df          SS        MS      F  p-value
──────────────────────────────────────────────────────
crew        3   2616.1986  872.0662  4.906    0.004
Residuals  76  13509.4136  177.7554
──────────────────────────────────────────────────────


Hypotheses (crew):
H₀: All group means are equal (no effect of crew)
H₁: At least one group mean differs (effect of crew exists)
Test statistic: F(3, 76) = 4.906
p-value: 0.004
Conclusion: Reject the null hypothesis (p = 0.004 < α = 0.050)
```

- Applying the Kruskal-Wallis,

```
baltimare_harbor |> kruskal_wallis(outcome = unload_minutes,
                                   grouping = crew)
```

```
Kruskal-Wallis test

M[Anchor Crew] = 5.44 (n = 20)
M[Driftwood Crew] = 13.67 (n = 20)
M[Saltmarsh Crew] = 5.09 (n = 20)
M[Tidewater Crew] = 9.475 (n = 20)

Hypotheses:
H₀: All group distributions are equal (no effect of crew)
H₁: At least one group distribution differs in location (effect of crew exists)
Test statistic: H(3) = 7.917
p-value: 0.048
Conclusion: Reject the null hypothesis (p = 0.048 < α = 0.050)
```

Q: Should we perform posthoc testing?

Q: Treat this as a confirmatory study and use the following output to determine which pairwise differences exist. Circle the output you are interpreting.

- Posthoc tests:

```
baltimare_harbor |> posthoc(outcome = unload_minutes,
                            grouping = crew,
                            adjust = TRUE,
                            nonparametric = FALSE)
```

```
Post-hoc analysis: Tukey's HSD

Following: one-way ANOVA (unload_minutes ~ crew)

Comparison                          Difference  p-value
──────────────────────────────────────────────────────
Driftwood Crew vs Anchor Crew         13.2555    0.012  *
Saltmarsh Crew vs Anchor Crew         -1.0660    0.994
Tidewater Crew vs Anchor Crew          1.9050    0.969
Saltmarsh Crew vs Driftwood Crew     -14.3215    0.006  *
Tidewater Crew vs Driftwood Crew     -11.3505    0.042  *
Tidewater Crew vs Saltmarsh Crew       2.9710    0.895
──────────────────────────────────────────────────────

* p < α = 0.050
```

```
baltimare_harbor |> posthoc(outcome = unload_minutes,
                            grouping = crew,
                            adjust = FALSE,
                            nonparametric = FALSE)
```

```
Post-hoc analysis: Fisher's LSD

Following: one-way ANOVA (unload_minutes ~ crew)

Comparison                          Difference  p-value
──────────────────────────────────────────────────────
Driftwood Crew vs Anchor Crew         13.2555    0.002  *
Saltmarsh Crew vs Anchor Crew         -1.0660    0.801
Saltmarsh Crew vs Driftwood Crew     -14.3215    0.001  *
Tidewater Crew vs Anchor Crew          1.9050    0.653
Tidewater Crew vs Driftwood Crew     -11.3505    0.009  *
Tidewater Crew vs Saltmarsh Crew       2.9710    0.483
──────────────────────────────────────────────────────

* p < α = 0.050
```

```
baltimare_harbor |> posthoc(outcome = unload_minutes,
                            grouping = crew,
                            adjust = TRUE,
                            nonparametric = TRUE)
```

```
Post-hoc analysis: Dunn's test (Bonferroni adjustment)

Following: Kruskal-Wallis (unload_minutes ~ crew)

Comparison                              z  p-value
──────────────────────────────────────────────────
Anchor Crew vs Driftwood Crew      -2.286    0.067
Anchor Crew vs Saltmarsh Crew       0.214    1.000
Driftwood Crew vs Saltmarsh Crew    2.501    0.037  *
Anchor Crew vs Tidewater Crew      -1.085    0.833
Driftwood Crew vs Tidewater Crew    1.201    0.689
Saltmarsh Crew vs Tidewater Crew   -1.300    0.581
──────────────────────────────────────────────────

* p < α = 0.050
```

```
baltimare_harbor |> posthoc(outcome = unload_minutes,
                            grouping = crew,
                            adjust = FALSE,
                            nonparametric = TRUE)
```

```
Post-hoc analysis: Dunn's test (no adjustment)

Following: Kruskal-Wallis (unload_minutes ~ crew)

Comparison                              z  p-value
──────────────────────────────────────────────────
Anchor Crew vs Driftwood Crew      -2.286    0.011  *
Anchor Crew vs Saltmarsh Crew       0.214    0.415
Driftwood Crew vs Saltmarsh Crew    2.501    0.006  *
Anchor Crew vs Tidewater Crew      -1.085    0.139
Driftwood Crew vs Tidewater Crew    1.201    0.115
Saltmarsh Crew vs Tidewater Crew   -1.300    0.097
──────────────────────────────────────────────────

* p < α = 0.050
```